AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

M251-M257
Object Oriented Programming Using Java

Final Examination
Spring Semester 2024/2025

Date: 28 / May / 2025

Number of Exam Pages: 4 (including this cover sheet)	Time Allowed: 3 hours

Instructions:

1. This exam consists of three parts. It is required to solve **ALL** of them in the external answer booklet:
   - **Part 1: 5** MCQ (10 marks)
   - **Part 2: 3** short answer questions (35 marks)
   - **Part 3: 2** problem solving questions (55 marks)
2. Go through all questions before you start answering them.
3. It is not allowed to use any external material.
4. It is not allowed to use any electronic device (calculators, mobiles, ...)
5. At the end of the examination:
   - Check that you have written **your name, student ID and section number** on the first page of your answer booklet
   - Place your question paper in the answer booklet and hand over them to the invigilator.

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PART 1: MCQ [10 Marks]: Each question carries a weight of 2 marks

Question 1: [10 marks] Read the following questions and choose the best answer.

1. Methods declared as _____ cannot be overridden.
   a. abstract b. final c. protected
   d. All the above e. None of the above

2. Which of the following does not have a constructor?
   a. concrete class b. abstract class c. interface
   d. A and B e. B and C

3. Which of the following is a checked Exception?
   a. EOFException b. ArithmeticException c. NullPointerException
   d. All the above e. None of the above

4. _____ places the nodes in in a single column.
   a. FlowPane b. HBox c. VBox
   d. BorderPane e. GridPane

5. _____ method moves the thread from the running state to the blocked state.
   a. sleep b. stop c. yield
   d. notify e. A and B

PART 2: Short Answer Questions [35 Marks]

Question 2: [12 marks: 2+2+2+2+4]

Given the following code, answer the given questions:

```
public interface Movable {
  public void move();
  public void stop();
}
```

```
public class PC implements Movable {
  public void move() {
    System.out.println("go");
  }
  public void stop() {
    System.out.println("wait");
  }
  public void printLocation() {
    System.out.println("Gaza");
  }
}
```

```
public class Test {
  public static void main(String[] a) {
    Movable m = new PC();
    PC p = new PC();
    // valid or not?
    //1// m.printLocation();
    //2// Movable x = new Movable();
    p.printLocation();
    p.move();
    m.stop();
    m.move();
  }
}
```

A. Draw a simple class diagram representing the relationship between `Movable` and `PC`.
B. Extract from the code an abstract method **and** mention the name of its class (or interface).
C. **Is** the code in the line that starts with the comment `//1//` **valid? Explain.**
D. **Is** the code in the line that starts with the comment `//2//` **valid? Explain.**
E. Provide the exact output of the class `Test`.

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Question 3: [13 marks]

Write the code of a method that takes a Set of words as an argument. The method should do the following:

- Declare and create an empty ArrayList of words.
- Iterate over all words in the Set and add only the words start by letter 'H' to the ArrayList.
- Print the words in the ArrayList in reverse order.

For example, if the Set = ["Hadi", "Nada", "Hisham", "Hana"] the method should create the ArrayList ["Hadi", "Hisham", "Hana"] and print on the screen **Hana Hisham Hadi**.

Question 4: [10 marks]

What does the following method return for **n = 12**? (Show your working steps in detail)

```
public static int check(int n) {
    if (n == 1) return 0;
    return 1 + check(n/2);
}
```

PART 3: Problem Solving Questions [55 Marks]

Question 5: [42 marks]

Introduction:
There are many universities in the country. Some universities have many branches. There is a director (who is a professor) for each branch. Those universities store the names of branches and all information about their directors in a Map.

1. Write the class **Professor** which includes: **[15 marks: 2+3+2+2+4+2]**
   a) 3 attributes: full name, email and age.
   b) a zero-arg constructor and a mutli-arg constructor.
   c) getters for the attributes **full name** and **age**.
   d) **toString()** method which overrides the Object's toString() method.
   e) **equals()** method which overrides the Object's equals() method. It is enough to compare the **Professor full name**.
   f) **compareTo()** method which implements the Comparable's compareTo() method. You should use the **Professor full name** attribute only in the comparison.

2. Write the class **University** which includes: (add the necessary **import** statements)
   [22 marks: 1+2+3+9+7]
   a) 2 attributes: University name and a **Map** called **directors** to store branches' names as keys (e.g. Kuwait) and **Professors** (the director of each branch) as values. The map should be sorted according to the keys (alphabetically).
   b) one-arg constructor that accepts the University name as an argument and creates a new empty Map to store the directors of the branches.
   c) method **addDirector()** that accepts 2 arguments: branch name and a **Professor** object. It adds this pair to the map **directors**.
   d) method **oldestDirector()** that returns the name of the oldest director in the map **directors** (whose age is greater than the age of all other directors).
   e) method **saveIntoFile()** which accepts a file name as an argument. It stores the University name in a separate line. Then stores each branch in a different line as follows:
   branch name: all information of the director of the branch.

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3. Write a code to test your classes by performing the following tasks: **[5 marks]**
   a) create a **University** object called **myUniversity** where the University name is **"AOU"**.
   b) create 2 **Professor**s and assign them as directors to 2 different branches in **myUniversity** by calling a suitable method.
   c) print the name of the oldest director in **myUniversity** by calling a suitable method.
   d) call the method **saveIntoFile()** to save the required information.

Question 6: [13 marks: 7 + 6]

Based on the classes of Part 3 - Question 5,
Given the following code which displays the following GUI application:

```
public void start(Stage primaryStage) {
 //Missing Code (1): Initialization

   University myUniversity = ..............

   Label directorNameL = ..............
   Label directorEmailL = ..............
   Label directorAgeL = ..............
   Label branchNameL = ..............

   TextField directorNameTF = ..............
   TextField directorEmailTF = ..............
   TextField directorAgeTF = ..............
   TextField branchNameTF = ..............

   Button addB = ..............

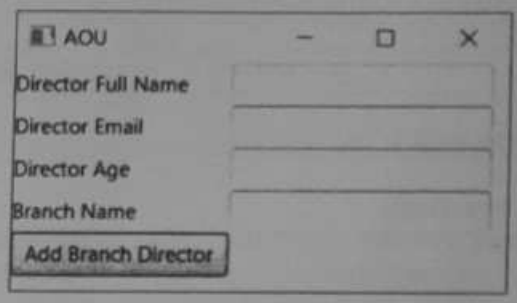
   GridPane pane = ..............
   primaryStage.setTitle(..............);

   pane.addColumn(0, directorNameL, ..............., ..............., ..............., ..............);
   pane.addColumn(1, directorNameTF, ..............., ..............., ..............);

   Scene scene = new Scene(..............);
   primaryStage.setScene(..............);

   primaryStage.show();

 //Missing Code (2): Clicking on the button
 ..............
 ..............
 ..............
}
```



1. Complete the first missing code to initialize the UI controls.
2. Complete the second missing code such that when the user fills in the required data and clicks the button **"Add Branch Director"**, a `Professor` object will be created (using the first 3 fields) and assigned as a branch director (in `myUniversity`) to the branch its name is in the fourth field.

End of Questions

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